TechRush Problem Statements

Front-End Development(Web/App)

1. Design/Redesign a Responsive Website

- Reimagine the user interface and experience of a well-known website.
- Ensure the website is fully responsive and user-friendly.
- Incorporate 3D elements and animations for an enhanced user experience.

2. Recipe Finder

- Create an application that helps users find recipes based on ingredients they have.
- Include features like recipe ratings, categories, and personalized recommendations.
- Ensure the application has a user-friendly interface and seamless navigation.

3. Personal Finance Dashboard

- Develop a dashboard for tracking personal finances, including expenses, income, and budgeting.
- Allow users to visualize their financial data with charts and graphs.
- Provide features for setting financial goals and tracking progress.

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Full-Stack Development

1. Photo Gallery Web Application

- **Concept**: Create a web-based photo gallery resembling Pinterest.
- **User Authentication**:
    - Implement account management.
    - Secure login system.
- **Photo Uploads**:
    - Enable users to upload photos with metadata (timestamps, captions).
    - Store photos in a database for efficient retrieval.
- **Homepage**:
    - Display all uploaded photos.
    - Implement pagination for easy browsing.
- **User Profiles**:
    - Allow users to view individual photo collections by clicking on user profiles.
- **Responsiveness**: Ensure the application is responsive on both desktop and mobile devices.
- **Documentation**: Document the implementation process comprehensively.

2. Basic E-commerce Website Prototype

- **Concept**: Develop a basic e-commerce website prototype.
- **User Authentication**:
    - Implement secure login for users.
- **Product Browsing**:
    - Allow users to browse products.
    - Add products to a virtual cart.
- **Simulated Purchases**:
    - Store selected items in a database upon clicking “buy”.
- **Product Management**:
    - Enable authenticated users to add new products.
- **Interactivity**:
    - Enhance interactivity by allowing product addition.
- **Responsiveness**: Ensure the prototype is responsive on various devices.
- **Instructions**: Provide clear instructions for setup and usage.
- **Focus**: Emphasize essential functionalities like product management and user interaction.
- **Note**: Do not implement actual payment processing.

3. Pokémon Pokedex Website

- **Concept**: Design or improve a web-based Pokémon Pokedex.
- **APIs**:
    - Use APIs like PokéAPI to enhance user interaction and information accessibility.
- **Search and Filter**:
    - Allow users to search Pokémon by name.
    - Filter Pokémon by type or attributes (abilities, stats).
- **Detailed Data**:
    - Fetch and display detailed data (abilities, moves, stats, evolutionary paths).
- **User Accounts**:
    - Implement user accounts for saving favorites.
- **Interactive Elements**:
    - Add animated sprites or habitat maps.
- **Community Features**:
    - Include comments and ratings.
- **User-Friendliness**: Ensure the application is user-friendly and intuitive.
- **Responsiveness**: Ensure responsiveness across various devices.
- **Documentation**: Provide comprehensive documentation of the application.

Artificial Intelligence/Machine Learning.

1. Diabetes Prediction

- Develop a Machine Learning model that can predict whether an individual has diabetes.
- Include various parameters critical for diagnosing diabetes.
- Utilize generative AI to provide personalized assistance to individuals who test positive.

2. Pre-owned Car Price Prediction

- Develop a Machine Learning model that predicts the correct price of a pre-owned car.
- Ensure the model uses an up-to-date dataset relevant to the Indian car market.
- Consider various factors that affect a car's price for accurate predictions.

3. Career Guidance Chatbot

- Develop a career guidance chatbot that assists students in making career choices.
- Collect necessary information about the student, including performance and field of interest.
- Provide guidance considering the scope in the chosen field and available job opportunities.